Stochastic Gradients & Stationarity

Ghadimi & Lan (2013) | *Stochastic First- and Zeroth-Order Methods for Nonconvex Stochastic Programming* | SIAM J. Optim.

Official paper: https://doi.org/10.1137/120880811 RQ2: Dynamics | RQ4: Theory vs practice

The paper in one minute

Question: If our objective is non-convex and our gradients are noisy, can an optimization algorithm still be guaranteed to make progress? **Answer:** Yes. The authors propose a *randomized stochastic gradient* (RSG) method and bound the work needed to find an *approximately stationary point*. This is *not* a guarantee of reaching a global or local minimum.

1. First principles: what does "stochastic" mean?

Stochastic = involving randomness. For a training loss averaged over n examples, full gradient descent uses every example: $f(w) = \frac{1}{n}\sum_{i=1}^{n} \ell_i(w)$, so $\nabla f(w) = \frac{1}{n}\sum_i \nabla \ell_i(w)$. SGD instead uses a gradient from a randomly sampled example or minibatch, written $G(w, \xi)$. Its update is

$$\boxed{w_{k+1} = w_k - \eta_k G(w_k, \xi_k)} \qquad \boxed{\mathbb{E}[G(w,\xi) \mid w] = \nabla f(w)}.$$

The expectation says the stochastic estimate is *unbiased*: each sample may point a little differently, but its average is the true gradient. The random variable ξ describes which data was sampled; η_k is the step size. Usually we also assume bounded noise, $\mathbb{E}\|G - \nabla f\|^2 \le \sigma^2$.

2. What does convergence mean for a non-convex loss?

A **stationary point** satisfies $\nabla f(w^*) = 0$. Because non-convex surfaces may have minima, maxima, and saddles, a small gradient alone cannot tell us which one we found. Therefore the target is often **approximate first-order stationarity**:

$$\boxed{\mathbb{E}\left[\|\nabla f(w_R)\|^2\right] \le \varepsilon.}$$

Here $\|\nabla f\|$ is the length of the gradient vector; ε is a chosen tolerance. RSG outputs a randomly selected iterate w_R from the optimization run (with probabilities set using its step-size schedule), rather than simply relying on the last iterate.

3. Why does the squared gradient become small? (compact math)

Assume f is L-**smooth**: its gradient does not change too sharply, $\|\nabla f(x) - \nabla f(y)\| \le L\|x - y\|$. This gives the standard descent inequality after one stochastic update:

$$f(w_{k+1}) \le f(w_k) - \eta\langle \nabla f(w_k), G_k\rangle + \frac{L\eta^2}{2}\|G_k\|^2.$$

Taking expectations, using the unbiased-gradient and bounded-variance assumptions, and taking $\eta \le 1/L$ yields

$$\frac{\eta}{2}\,\mathbb{E}\|\nabla f(w_k)\|^2 \le \mathbb{E}[f(w_k) - f(w_{k+1})] + \frac{L\eta^2\sigma^2}{2}.$$

Sum over N updates (the losses *telescope*), and let $\Delta = f(w_1) - f_{\inf} \ge 0$:

$$\boxed{\frac{1}{N}\sum_{k=1}^{N}\mathbb{E}\|\nabla f(w_k)\|^2 \le \frac{2\Delta}{\eta N} + L\eta\sigma^2.}$$

Meaning: More iterations reduce the first term; reducing the step size controls the noise term. With a suitable $\eta \propto N^{-1/2}$, the bound scales like $O(N^{-1/2})$ when gradient noise is present. This is a *teaching derivation of the standard smooth-SGD bound*; the paper gives its own RSG-specific output rules and complexity results.

Why it matters for our survey

First-order convergence is not saddle escape. Ghadimi–Lan establish that we can approach points with small gradients despite noisy updates. Later saddle-escape papers ask the harder question: can we additionally rule out *negative curvature* and avoid strict saddles? **Assumptions:** smoothness, suitable step sizes, unbiased controlled-noise gradients, and an objective bounded below. The paper also analyzes *zeroth-order* methods (using function values without direct gradients).

Our takeaway: SGD can be theoretically reliable at getting the gradient close to zero, even for non-convex losses. The catch is that "gradient near zero" does not necessarily mean "good minimum."